

20260904 AP Unit1 categorical data

1. A market researcher asked a group of men and women to choose their favorite color design from a sample of advertisements. The results are shown in the following table.

	Red with Black	Green with Black	Yellow with Black	Red with Blue	Green with Blue	Yellow with Blue
Men	21	15	8	12	35	9
Women	15	3	11	31	24	16

Which of the following statements is not supported by the table?

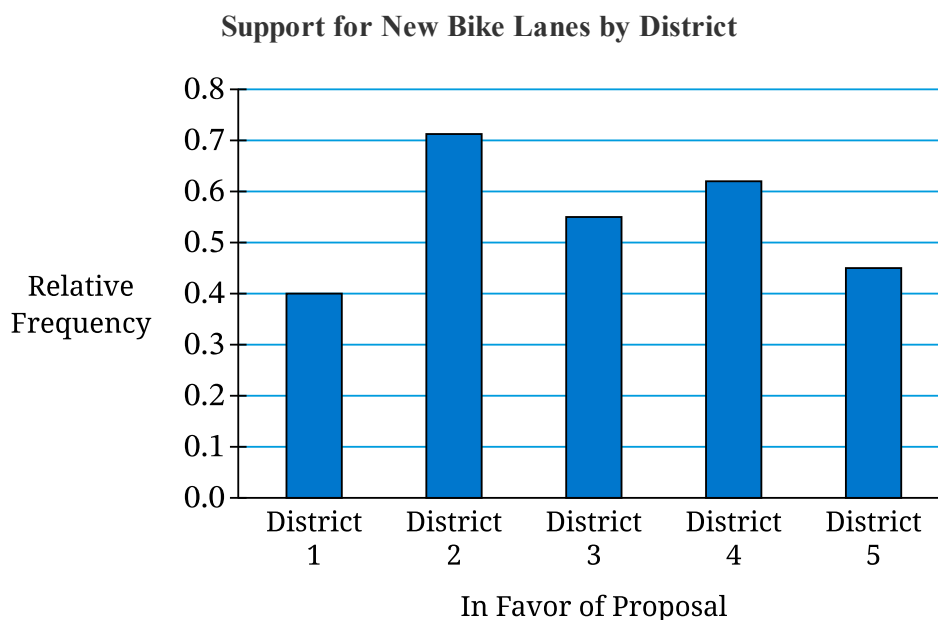
- (A) More women than men chose the color design yellow with black.
- (B) For men, the number who chose a design with black was greater than the number who chose a design with blue. ✓
- (C) The color design chosen by the most people was green with blue.
- (D) The total number of men surveyed by the market researcher was equal to the total number of women surveyed by the market researcher.

Answer B

Correct. The statement is not supported. The number who chose a design with black was 44, but 56 chose a design with blue.

20260904 AP Unit1 categorical data

2. Residents from five districts of a city were asked whether they were in favor of a city proposal to create new bike lanes on the roads. The bar chart shows the relative frequency for each district of responses that were in favor of the proposal.



Based on the chart, which of the following statements about the districts is supported by the data?

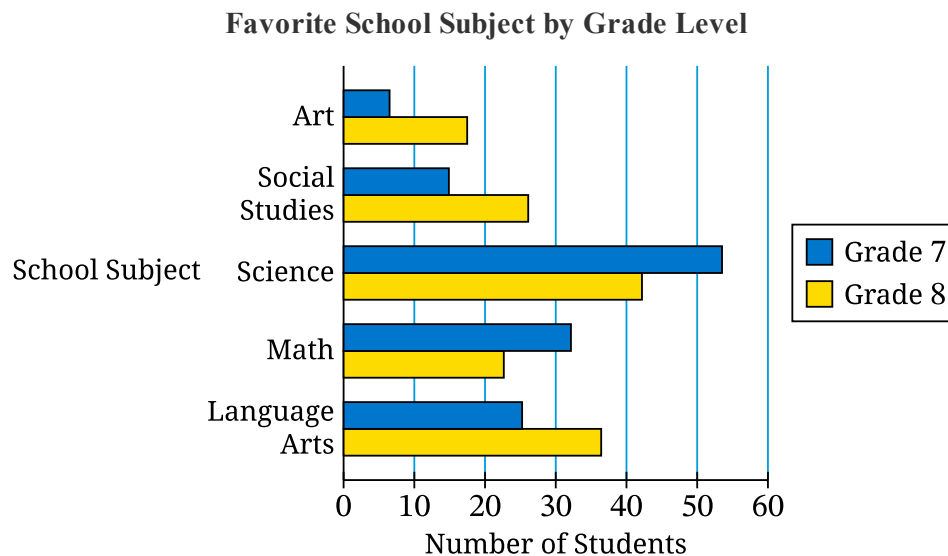
- (A) District 1 has the fewest number of residents who were in favor of the proposal.
- (B) District 2 has the greatest percentage of residents who were in favor of the proposal. ✓
- (C) In three districts, fewer than half of the residents who were in favor of the proposal.
- (D) All districts show at least 60% of residents in favor of the proposal.

Answer B

Correct. The bars represent percentages, and District 2 has the tallest bar (around 71%).

20260904 AP Unit1 categorical data

3. Two different groups of students, in grades 7 and 8, were asked which school subject was their favorite. The bar chart summarizes the responses by grade.



Based on the bar chart, which of the following statements about favorite school subjects is supported by the data?

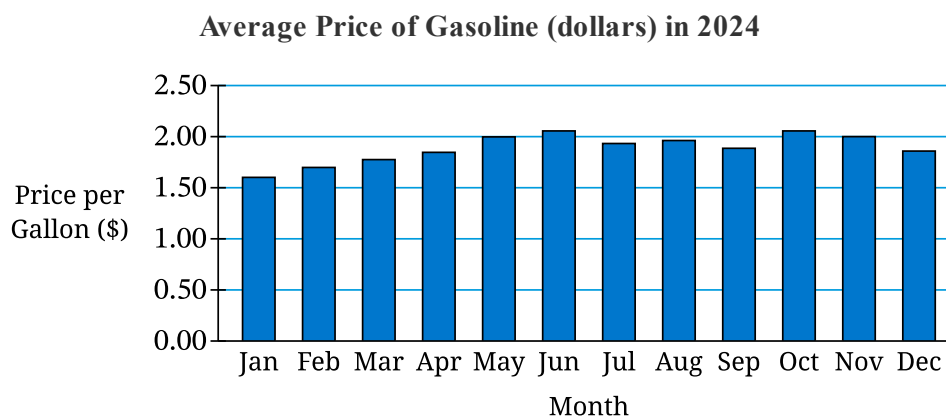
- (A) For students who chose social studies, the number of grade 8 students was at least 100% greater than the number of grade 7 students.
- (B) For students who chose art, the number of grade 8 students was at least 100% greater than the number of grade 7 students. ✓
- (C) For students who chose science, the number of grade 8 students was at least 50% less than the number of grade 7 students.
- (D) For students who chose math, the number of grade 8 students was at least 50% less than the number of grade 7 students.

Answer B

Correct. The grade 7 bar extends to at most 8 students. A 100% increase would reach 16 students, and the grade 8 bar extends to at least 16 students.

20260904 AP Unit1 categorical data

4. The following bar chart shows the average price per gallon of gasoline for each month in 2024.



Based on the bar chart, which of the following claims is not supported by the data?

- (A) The average gas price increased each month from January through June.
- (B) The gas price for every month was greater than \$1.50 per gallon.
- (C) The gas price was lowest in February. ✓
- (D) The September gas price was greater than in January.

Answer C

Correct. The statement is not supported since January, not February, has the lowest bar in the bar chart.

5. Data were collected on 35 students in a college math course. Which of the following is not an appropriate variable for the data set representing the 35 students?
- (A) Student birth month
 - (B) Student age
 - (C) Student address
 - (D) Number of students in the data set ✓

Answer D

Correct. A variable is a characteristic that can vary from student to student. The number of students in the data set is a single, fixed value (35).

20260904 AP Unit1 categorical data

6. The individual pieces of a popular candy come in four colors: red, blue, green, and white. In a box of 20 pieces, there were 8 red pieces.

Which of the following tables best displays relative frequency data on the colors of the candy pieces in the box?

20260904 AP Unit1 categorical data

(A)

Color	Relative Frequency
Red	8
Blue	5
Green	3
White	4

(B)

Color	Relative Frequency
Red	0.40
Blue	0.15
Green	0.30
White	0.15



20260904 AP Unit1 categorical data

(C)

Color	Relative Frequency
Red	0.25
Blue	0.25
Green	0.25
White	0.25

(D)

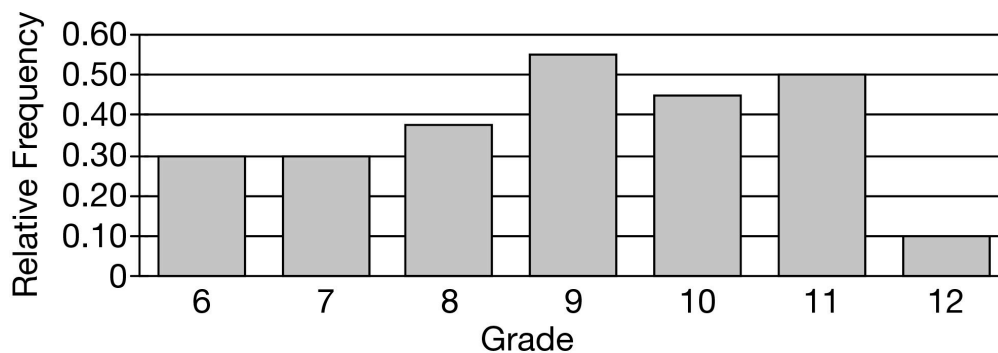
Color	Relative Frequency
Red	0.40
Blue	0.20
Green	0.25
White	0.05

Answer B

20260904 AP Unit1 categorical data

Correct. The relative frequency of red pieces is $\frac{8}{20} = 0.40$. All other relative frequencies are positive. The four relative frequencies add to 1.

7. In a certain school district, students from grade 6 through grade 12 can participate in a school-sponsored community service activity. The following bar chart shows the relative frequencies of students from each grade who participate in the community service activity.



Which of the following statements is supported by the bar chart?

- (A) The greatest number of participating students was in grade 9.
- (B) The relative frequency of all participating students in grades 6 and 7 combined was 0.60.
- (C) Grade 12 had the least relative frequency of participating students. ✓
- (D) Grade 11 had the greatest relative frequency of participating students.

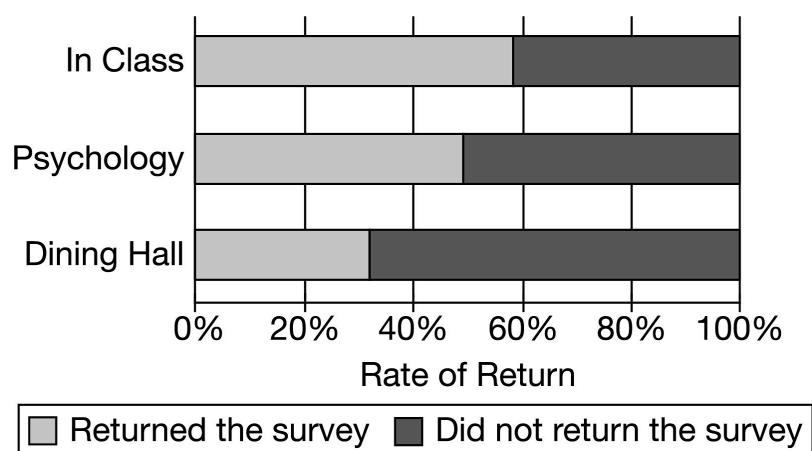
Answer C

Correct. The least relative frequency on the bar chart is 0.10, which corresponds to grade 12.

20260904 AP Unit1 categorical data

8. College researchers wanted to know under what conditions people are more likely to complete and return a survey. As part of a study, the researchers prepared three sets of identical surveys and used three methods of delivering and returning the surveys. The methods are described as follows.
- In Class: The surveys were given to students in a class, and students were asked to return completed surveys to their instructor.
 - Psychology: The surveys were given to students participating in a psychology experiment, and students were asked to return completed surveys to a collection box in the hallway of the psychology building.
 - Dining Hall: The surveys were given to students in the dining hall, and students were asked to return completed surveys to a collection box outside the dining hall.

The graph shows the percent of surveys returned and not returned for each delivery method.



Which statement about delivery method and rate of survey return is supported by the graph?

- (A) There is a positive association between delivery method and rate of return.
- (B) The number of surveys given using the Dining Hall delivery method was less than the number given using either of the other delivery methods.
- (C) The Psychology delivery method displays the most symmetric results; the other delivery methods display skewed results.
- (D) The In Class delivery method had the greatest rate of return, and the Dining Hall delivery method had the least rate of return. ✓

Answer D

Correct. According to the graph, the rate of return for the Dining Hall delivery method was approximately 33 percent, for the Psychology delivery method was approximately 48 percent, and for the In Class delivery method was approximately 58 percent. The In Class delivery method had the greatest rate of return, and the Dining Hall delivery method had the least rate of return.

20260904 AP Unit1 categorical data

9. Which of the following describes a continuous variable?
- (A) The number of apps downloaded from a website one day
 - (B) The diameters of the tree trunks at an evergreen farm ✓
 - (C) The number of baskets made by a basketball player
 - (D) The shoe sizes of all shoes on sale at a department store

Answer B

Correct. Diameter is a measure of length, and length is continuous because no matter how small the difference is between any two lengths, another fractional length exists between those lengths.

20260904 AP Unit1 categorical data

10. The relative frequency table shows reasons given by high school students for their most recent absence from school.

Reasons for Absences

Reason	Relative Frequency
Illness	0.57
Overslept	0.20
College visit	0.16
Did not finish assignment	0.07

Which of the following claims cannot be justified based on the data in the table?

- (A) More than one-third of the students were absent because of “Overslept” or “College visit.”
- (B) The reason given the fewest number of times was “Did not finish assignment.”
- (C) Only seven students were absent because of “Did not finish assignment.” ✓
- (D) Less than one-fourth of the students were absent because of “Overslept.”

Answer C

Correct. The statement is not supported by the data in the relative frequency table. The proportion of students who gave a reason of “Did not finish assignment” is 0.07, meaning that 7% of the total number of students, T , gave this reason. The number of students giving this reason is equal to $0.07 \times T$, but this number cannot be determined because T is not known.

20260904 AP Unit1 categorical data

11. A group of students were asked to name their favorite color. The responses are shown in the frequency table.

Students' Favorite Color

Color	Frequency
Red	15
Blue	30
Green	8
Yellow	12
Purple	25

Which of the following claims is supported by the data in the table?

- (A) More students chose purple than any other color.
- (B) Twice as many students chose blue as chose red. ✓
- (C) The total number of student responses is 80.
- (D) The combined number of students who chose green and yellow is greater than the number of students who chose purple.

Answer B

Correct. The number of students who chose red is 15, and 30 students chose blue. Because 30 is twice 15, the statement “Twice as many students chose blue as chose red” is supported by the data in the table.

20260904 AP Unit1 categorical data

12. Data will be collected on the following variables. Which variable can be considered discrete?

- (A) The height of a person
- (B) The length of a person's arm span
- (C) The time it takes for a person to solve a puzzle
- (D) The number of books a person finished reading last month

**Answer D**

Correct. The number of books is a counting number which, like all whole numbers, is discrete.

13. A child psychologist asked 100 five year olds and 50 ten year olds to name their favorite color. Their results are shown in the following table.

	Red	Orange	Yellow	Green	Blue	Purple
Five year olds	21	15	8	10	35	11
Ten year olds	7	2	6	15	12	8

Which of the following statements is supported by the table?

- (A) The percentage of five year olds who selected red or blue as their favorite color is greater than the percentage of ten year olds who selected red or blue as their favorite color.
- (B) The percentage of five year olds who selected yellow as their favorite color is greater than the percentage of ten year olds who selected yellow as their favorite color.
- (C) The percentage of children who selected red, yellow, or blue as their favorite color was equal for both ages.
- (D) Less than half of the ten year olds selected red, yellow, or blue as their favorite color.

**Answer A**

Correct. The statement is supported by the table. The percentage of five year olds who selected red or blue as their favorite color is $\frac{21+35}{100} = 56\%$, while the percentage of ten year olds who selected red or blue as their favorite color is $\frac{7+12}{50} = 38\%$.

20260904 AP Unit1 categorical data

14. The following frequency table shows the responses from a group of college students who were asked to choose their favorite flavor of ice cream.

Flavor	Frequency
Vanilla	100
Chocolate	65
Strawberry	35
Mint chip	45
Coffee	30
Butter pecan	25

Which of the following statements is not supported by the table?

- (A) One-third of the students chose vanilla.
(B) One-third of the students chose chocolate or strawberry.
(C) One-fourth of the students chose mint chip or coffee.
(D) One-half of the students chose vanilla or chocolate. ✓

Answer D

Correct. The total frequency of students is $100 + 65 + 35 + 45 + 30 + 25 = 300$. One-half of the students is $\frac{1}{2}(300) = 150$. The number of students who chose vanilla or chocolate is $100 + 65 = 165$. Since $165 \neq 150$, the statement “One-half of the students chose vanilla or chocolate” is not supported by the table.

20260904 AP Unit1 categorical data

15. Students in a class were asked to choose their favorite school subject. The following frequency table summarizes the responses.

School Subject	Frequency
Math	7
English	5
Science	4
Social Studies	2
Other	2

Which of the following statements is supported by the frequency table?

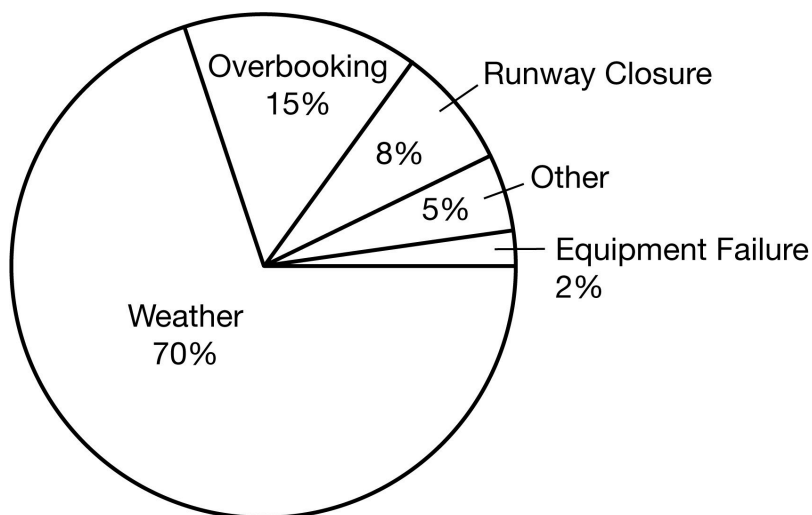
- (A) Fewer students chose math as their favorite subject than the combined number of students who chose science and social studies.
- (B) The total number of students in the class is less than 20.
- (C) Seventy percent of the students chose math as their favorite subject.
- (D) Less than half of the students chose math as their favorite subject. ✓

Answer D

Correct. Thirty-five percent of the students chose math as their favorite subject.

20260904 AP Unit1 categorical data

16. The following pie chart summarizes the results of a survey given to airlines about the primary reason for flight delays.



Which of the following statements is supported by the pie chart?

- (A) The reason given most frequently was runway closure.
- (B) More delays were caused by weather than by all other reasons combined. ✓
- (C) More delays were caused by runway closure than were caused by overbooking.
- (D) Overbooking and runway closure accounted for greater than one-fourth of the reasons given for flight delays.

Answer B

Correct. The percentage for weather is greater than 50%.

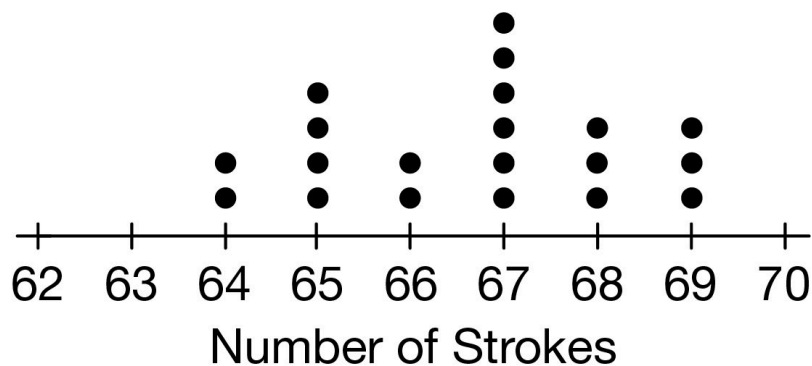
17. Staff members of a high school newspaper want to obtain an estimate of the average number of years teachers in the state have been teaching. At an educational conference attended by many teachers in the state, the staff members randomly selected 50 conference attendees and asked the attendees how long they have been teaching.

Which of the following describes the sample and the population to which it would be most reasonable for the staff members to generalize the results?

- (A) The sample is the 50 conference attendees, and the population is all teachers in the state.
- (B) The sample is the 50 conference attendees, and the population is all conference attendees. ✓
- (C) The sample is all conference attendees, and the population is all teachers in the state.
- (D) The sample is all conference attendees, and the population is the 50 conference attendees.

20260904 AP Unit1 categorical data

18. The following dotplot shows the number of strokes taken by 20 professional golfers competing at an 18-hole golf course.



Which of the following best describes the type of variable represented in the dotplot?

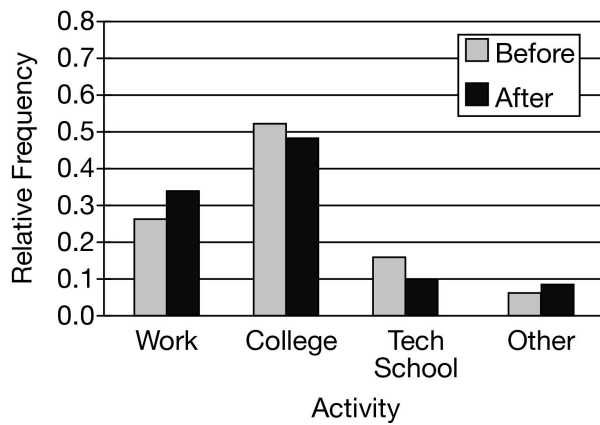
- (A) Quantitative and discrete
- (B) Quantitative and continuous
- (C) Categorical and discrete
- (D) Quantitative and categorical

**Answer A**

Correct. The number of strokes is a whole number.

20260904 AP Unit1 categorical data

19. Just before graduation, a random sample of high school seniors were asked what their plans were for the coming year. A year after graduation, those same individuals were asked what their main activity was in the year since graduation. Their responses are shown in the following bar chart.



Which of the following statements is supported by the bar chart?

- (A) The percentage of students who selected Work as their main activity increased after graduation. ✓
- (B) The percentage of students who selected College as their main activity increased after graduation.
- (C) Before graduation, the percentage of students who selected Tech School as their main activity was greater than the percentage who selected Work as their main activity.
- (D) After graduation, the percentage of students who selected College as their main activity was twice the percentage of those who selected Work as their main activity.

Answer A

Correct. The bars for Work increased from below 0.3 to above 0.3.

20260904 AP Unit1 categorical data

20. A commercial tree farmer grows Scotch pine trees in a large tree grove. Every year, the farmer harvests trees, and plants new trees. Three years ago, the average height of one-year-old Scotch pine trees in the grove was 32 centimeters (cm). The farmer has been experimenting with a new irrigation system in a section of the grove to increase the height of the pine trees. The farmer selects a random sample of 40 one-year-old Scotch pine trees from this section of the grove and measures the height of each tree. If the data provide convincing statistical evidence that the mean height is greater than 32 cm, the farmer will expand the irrigation system to the entire grove.

Which of the following is the correct investigative question for this statistical study?

- (A) What is the height of a randomly selected one-year-old Scotch pine tree from the grove?
- (B) For all one-year-old Scotch pine trees in this section of the grove, is the mean height different than 32 cm?
- (C) For all Scotch pine trees from the grove, is the mean height greater than 32 cm?
- (D) For all one-year-old Scotch pine trees in this section of the grove, is the mean height greater than 32 cm? ✓

Answer D

Correct. The population is correct (all one-year-old Scotch pine trees in this section of the grove). The direction of the hypothesis test is correct (greater than 32 cm).